
EXPLOR
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EVOLV
National Level
Student Technical Paper & Project Contest and Exhibition
STEPCONE
16th Edition
FEB 16 & 17 2024
VIKSIT BHARAT
2K47
TECHNICAL
PAPER PRESENTATION
Total Prizes Worth
21000
Registration Fee :
200/-
SCAN ME
ORGANIZING COMMITTEE
Faculty Coordinators :
Dr.S.Chiranjeeva Rao
Dr.Amol R Bute
Students Coordinators :
K.Siva Sai Charan - 8790976981
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stepone 2024

TECHNICAL PAPER PRESENTATION - 2024



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2. **GUIDELINES FOR PAPER PRESENTATION & SUBMISSION** - **01**

3. **TECHNICAL PAPER PRESENTATION THEMES** - **03**

3.1. **Computer Science and Engineering(including AIDS&AIML)** - **06**

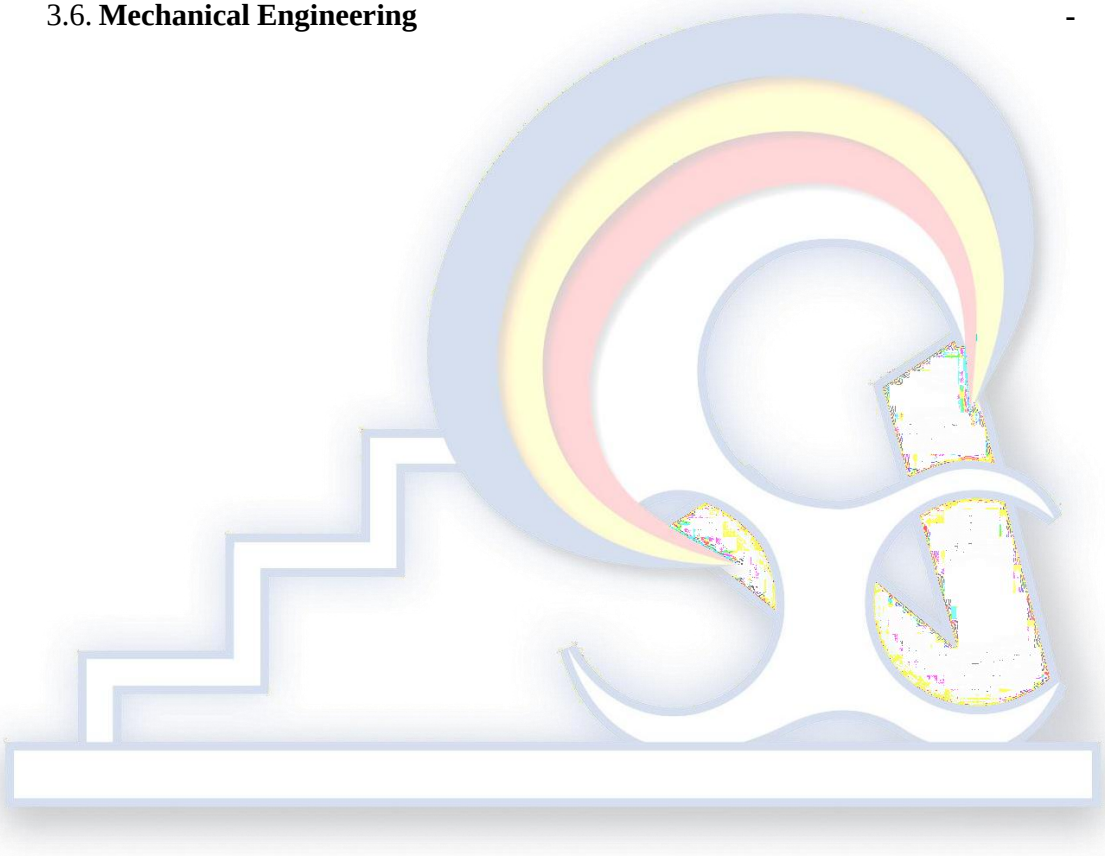
3.2. **Information Technology** - **07**

3.3. **Electronics and Communication Engineering** - **08**

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3.5. **Civil Engineering** - **10**

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TECHNICAL PAPER PRESENTATION - 2024

The STEPCONE platform aims to showcase empirical research outcomes across diverse engineering domains, emphasizing the identification of research gaps and advancements in emerging areas of engineering and technology. The platform also encourages the examination of current scientific contributions relevant to engineering research. Research communication, encompassing both novel perspectives on traditional domains and cutting-edge technologies, is highly encouraged.

We invite full papers in the prescribed format from enthusiastic student researchers. Submissions are accepted either as a single-author contribution or as a collaborative effort with a maximum of two authors.

We look forward to receiving innovative research contributions that contribute to the dialogue on engineering advancements and further enrich the STEPCONE platform with valuable insights.

1. DETAILS

Registration fee per author: Rs.200/-

Maximum no of authors: 2 per paper

1. There is no basic registration fee. Only event registration fee

2. Participation certificates will be given to all

3. 20 % of papers will be published in journals with ISSN Number .

2. GUIDELINES FOR PAPER PRESENTATION & SUBMISSION

A. Submission Guidelines:

1. **Abstract Length:** Please submit an abstract of 100-150 words, providing a concise overview of your research paper.
2. **Submission Deadline:** The deadline for abstract submission is **February 5th, 2024.**
3. **Intimation of acceptance : on or before within 12 working hours after submission.**

Full Paper Submission:

3. **Selected Candidates:** Those whose abstracts are shortlisted will be notified to submit the full paper.

4. **Full Paper Length:** The full paper should be between 2000-5000 words. IEEE format (Citation and Reference) should be submitted in hard copy to the organizers and the soft copy should be submitted at https://docs.google.com/forms/d/e/1FAIpQLSfIgvUZBW-tBrG0sUo66Kxx3lTk4nsCCUCVmJGcbkbg4kj9Lg/viewform?usp=sf_link

   Paper should be typed in "Times New Roman" font size "12" and for heading font size of "14" in bold.

   Paper contents should be arranged in the following order but can be modified if discipline/nature of work demands:

   – Title

   – Abstract

   – Keywords

   – Introduction

   – Materials & Methods

   – Result and Discussion

   – Conclusion

   – References

5. **Submission Deadline:** Full papers must be submitted by **February 11th , 2024**, for consideration.

6. **Author Details:** Please include the following details in the order in your paper:

   - serial order for all authors
   - Current Year of Study
   - Branch
   - Institute Name and Address
   - E-Mail ID
   - Phone Number

7. **Corresponding Author:** One author should submit on behalf of the entire group and will be treated as the Corresponding Author.

Presentation:

8. **Presentation Duration:** Each participant should be prepared for a 10-minute presentation of their paper at the venue. (**need to share the presentation with paper ID as file name before the 14th Feb 2024**)
9. **Schedule:** The presentation schedule will be communicated in the reply email after the abstract selection process.

We encourage you to participate actively in this event and look forward to receiving your abstracts. Should you have any queries, feel free to reach out to us at [technicalpaper2024@gmail.com].



3. TECHNICAL PAPER PRESENTATION THEMES

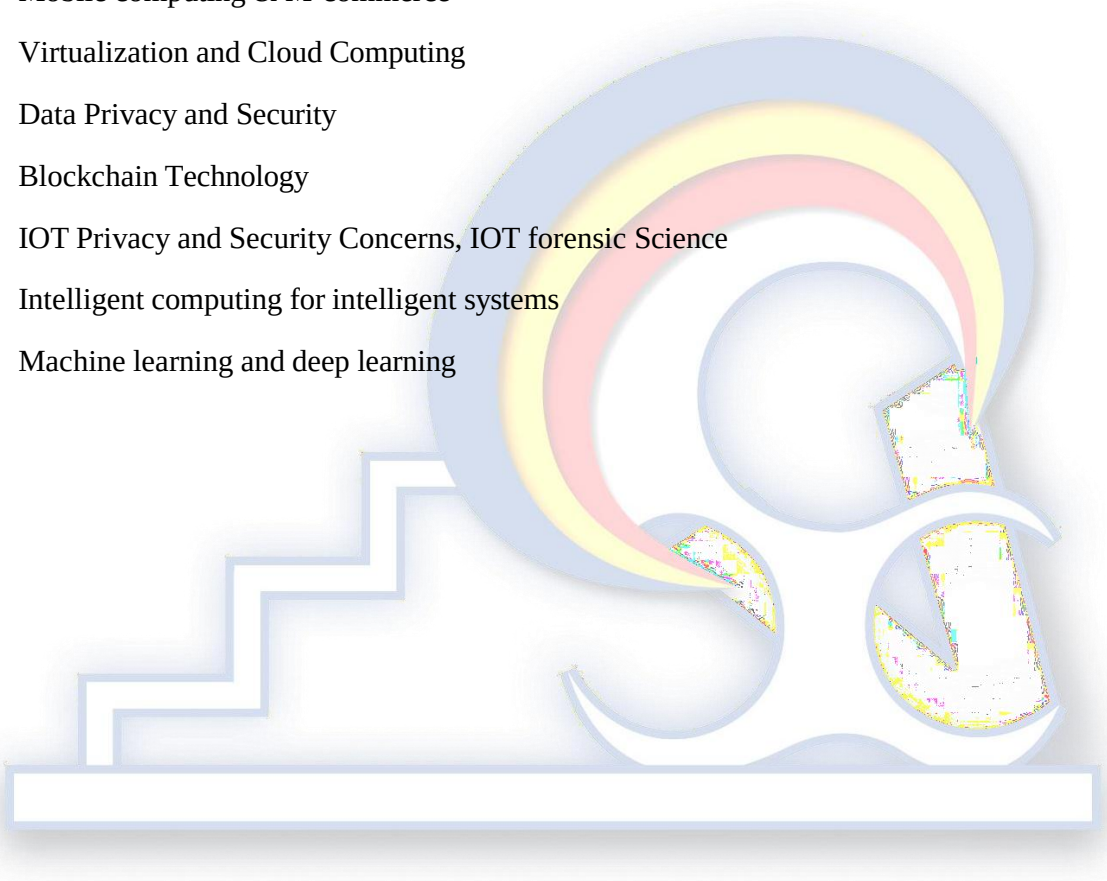
3.1. Computer Science and Engineering, Artificial intelligence and machine learning, Artificial intelligence and Data Science.

1. Cyber Security
2. Blockchain
3. AI & ML – Recent Trends and Applications
4. DevOps in Software Engineering
5. AI in Data Science
6. Wireless Sensor Networks
7. Metaverse
8. Quantum computing
9. Natural Language Processing(NLP)
10. IOT-Based Security Mode
11. Augmented Reality/Virtual Reality
12. Quantum Computing/High Performance Computing
13. Holographic Data Storage
14. Cloud Computing-Ensuring Data Security
15. Neuromorphic Computing
16. IDMA – The Future of Wireless Technology
17. 5G Technology and Its Implications
18. Exoskeletons in Rehabilitation and Industry
19. Web Technologies
20. Latest trends in Cyber Security

Ms.K. Jyotsna	Dr. M. Satish	Mr.CH.Sekhar	Dr. A. V. Ramana	Dr.K. Srividya
Student I/C	**Faculty SPOC-CSE**	**Faculty SPOC-AI&DS/ML**	**HOD-CSE**	**HOD-AI&DS/ML**

Information Technology

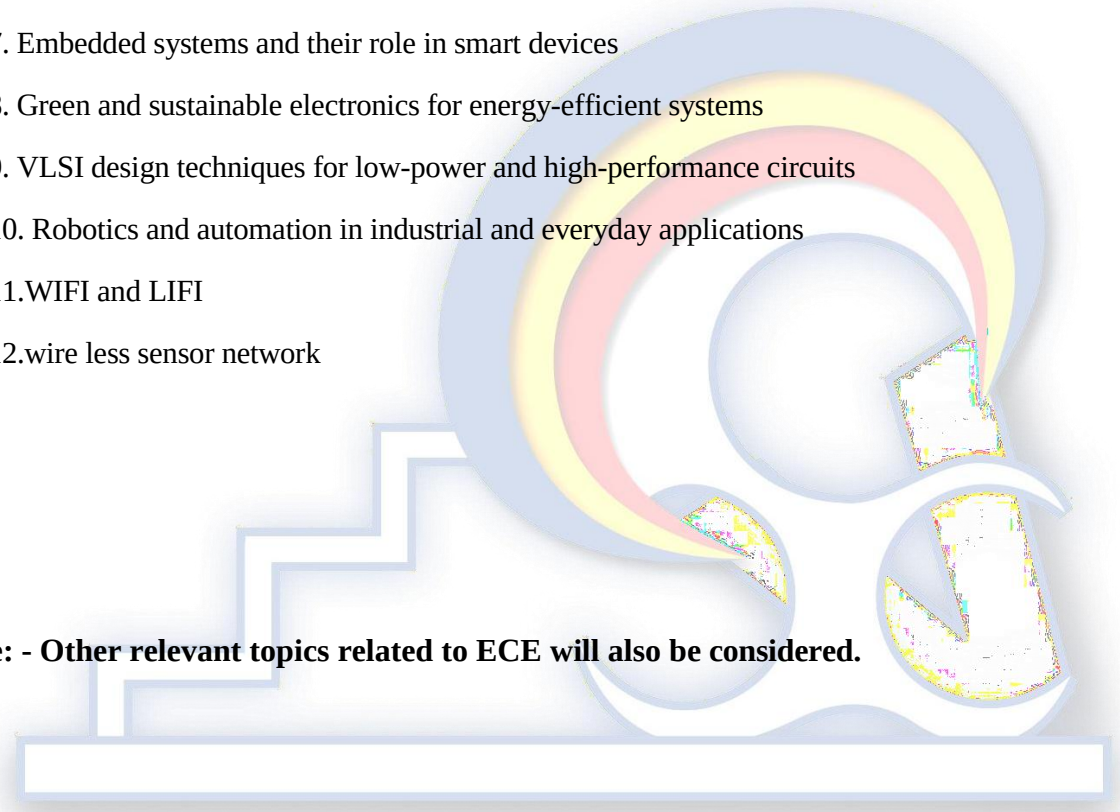
1. Machine learning algorithms
2. Predictive modelling
3. Computational complexity and optimization
4. Next-Generation Communication Technologies
5. Wireless Sensor Networks
6. Mobile computing & M-commerce
7. Virtualization and Cloud Computing
8. Data Privacy and Security
9. Blockchain Technology
10. IOT Privacy and Security Concerns, IOT forensic Science
11. Intelligent computing for intelligent systems
12. Machine learning and deep learning



Mr. K.Siva Sai Charan	Mr. Ch. Anil Kumar	Dr.V.Vasudha Rani
Student I/C	**Faculty SPOC**	**HOD - IT**

Electronics and Communication Engineering

1. Internet of Things (IoT) and its applications in various fields
2. 2. Artificial Intelligence (AI) and Machine Learning (ML) in signal processing, agriculture and other applications
3. 3. 6G and 5G technology its impact on communication systems
4. 4. Advanced antenna systems for wireless communication
5. 5. Image and video processing techniques for computer vision applications
6. 6. Biomedical signal processing and its applications in healthcare
7. 7. Embedded systems and their role in smart devices
8. 8. Green and sustainable electronics for energy-efficient systems
9. 9. VLSI design techniques for low-power and high-performance circuits
10. 10. Robotics and automation in industrial and everyday applications
11. 11.WIFI and LIFI
12. 12.wire less sensor network

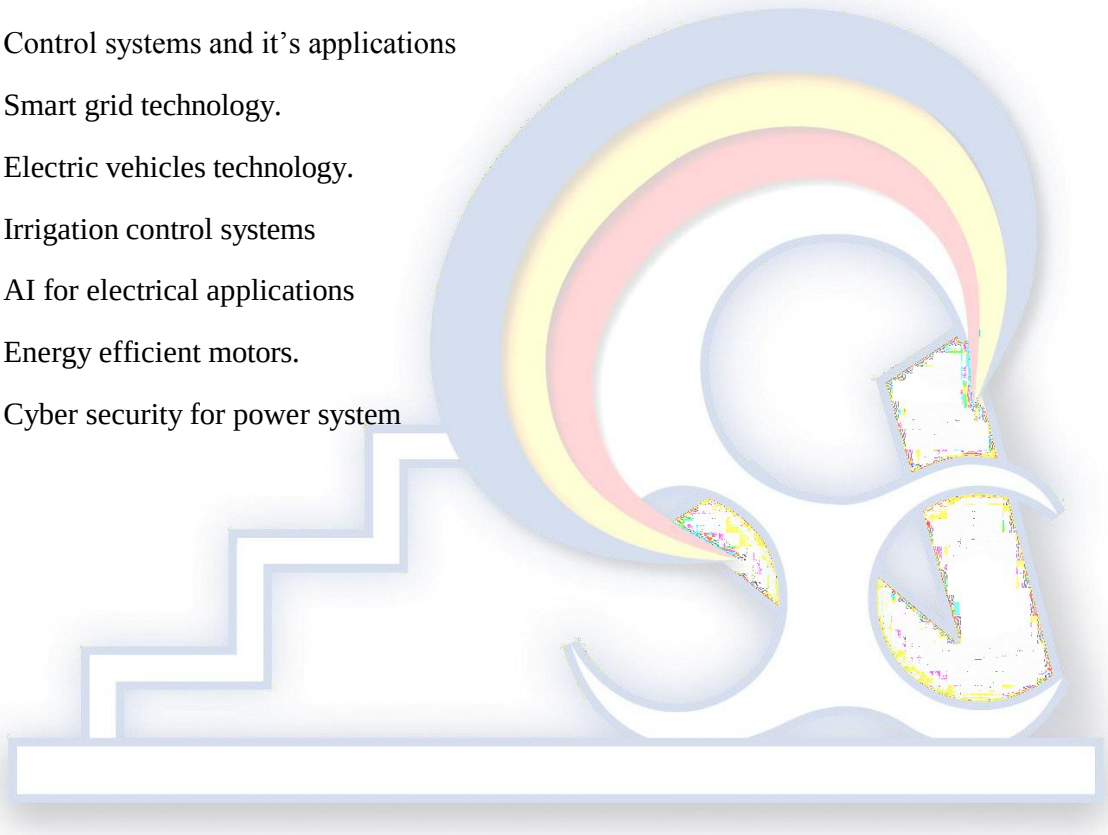


Note: - Other relevant topics related to ECE will also be considered.

Mr.S.Venu	Dr.A.Siva Sangari	Dr.V.Jagan Naveen
Student I/C	**Faculty SPOC**	**HOD - ECE**

Electrical and Electronics Engineering

1. Renewable energy sources.
2. Battery management systems
3. Power electronics and it's applications
4. Power systems and it's applications
5. Wireless electricity
6. Health monitoring devices
7. IOT based home appliances
8. Control systems and it's applications
9. Smart grid technology.
10. Electric vehicles technology.
11. Irrigation control systems
12. AI for electrical applications
13. Energy efficient motors.
14. Cyber security for power system



Ms. I.Vidhyadhari	Dr. J S V Shiva Kumar	Dr.P. Ramana
Student I/C	**Faculty SPOC**	**HOD - EEE**

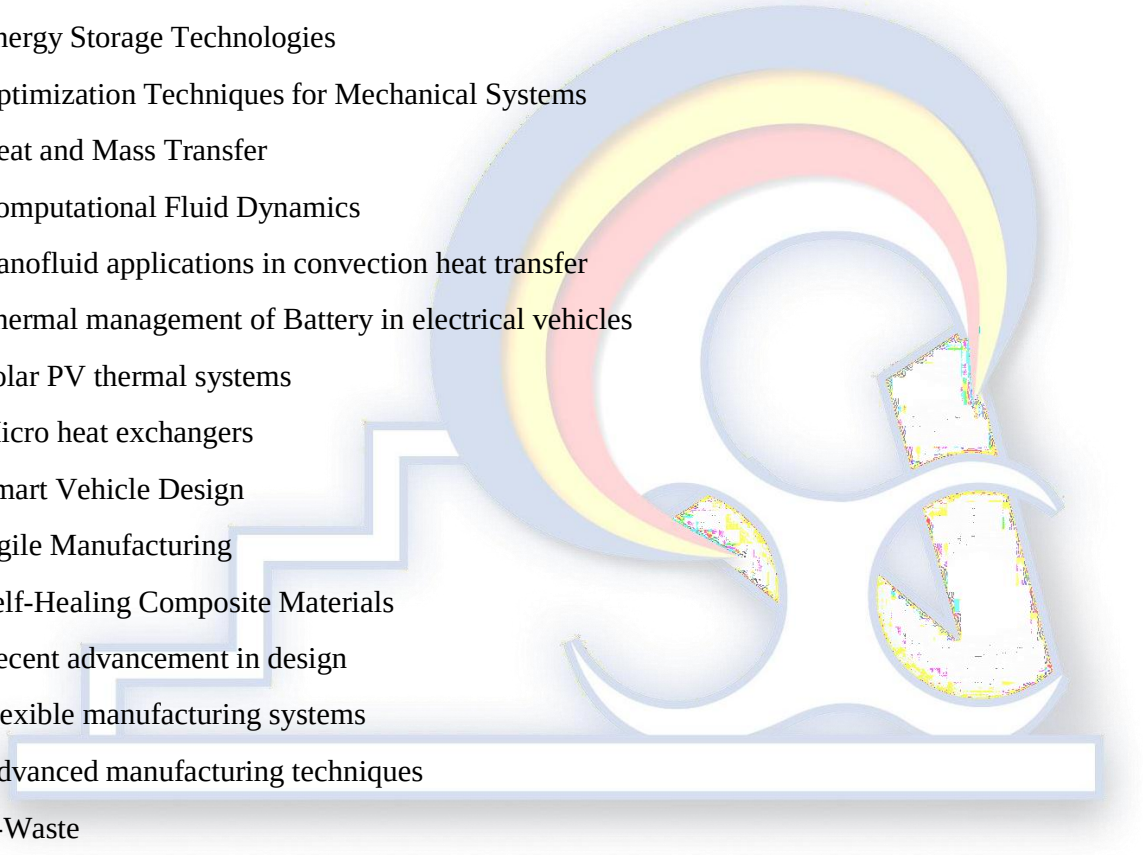
Civil Engineering

1. EBIM applications in structural engineering
2. New and advanced construction materials
3. Advanced Water proofing materials in structural applications
4. Building for Resilience: Designing Infrastructure to Withstand Natural Disasters
5. Building optimal design
6. Fossil fuels and renewable energy
7. Oil spills: Environmental Impact, Human Impact, Preventative Measures
8. Desalination:
9. Soil pollutions and treatment
10. Phytoremediation
11. Microbial Fuel Cells for Wastewater Treatment
12. Remote Sensing Application in Environmental Protection
13. Sustainable Water Management System
14. Air Quality Prediction using Machine Learning
15. Applications of IOT in Solid Waste Management
16. Marine geotechnics
17. Dams and tunnelling
18. Geosynthetics in soil stabilization
19. Geohazards and Risk Assessment
20. Sesimic control
21. Soil Dynamics
22. Soil Structure Interaction
23. Unsaturated Soil Mechanics
24. Unconventional materials and methods in soil stabilization
25. Liners in Municipal solid waste management
26. Sustainable Materials for pavement Design
27. Applications in Transportation Engineering
28. Intelligent Transport Systems
29. Asphalt products and properties

Ms.G.Lavanya	Mr. K. Naga Rajesh	Dr. G. Ganesh Prabhu
Student I/C	**Faculty SPOC**	**HOD - CIVIL**

Mechanical Engineering

1. Green Technologies
2. Sustainable Manufacture
3. Robotic and Human Machine Affairs
4. Industry 4.0
5. Energy Storage Technologies
6. Optimization Techniques for Mechanical Systems
7. Heat and Mass Transfer
8. Computational Fluid Dynamics
9. Nanofluid applications in convection heat transfer
10. Thermal management of Battery in electrical vehicles
11. Solar PV thermal systems
12. Micro heat exchangers
13. Smart Vehicle Design
14. Agile Manufacturing
15. Self-Healing Composite Materials
16. Recent advancement in design
17. Flexible manufacturing systems
18. Advanced manufacturing techniques
19. E-Waste
20. Machine learning applications in mechanical engineering



Topics related to contemporary engineering fields are also considered

Ms.K.Jasmitha Sai	Dr. S.Ravi Babu	Dr. G. Sasi Kumar
Student I/C	**Faculty SPOC**	**HOD - MECH**

FACULTY -COORDINATORS:

Dr.S.Chiranjeeva Rao-9849748090

Dr.Amol R Bute-7709968675

Co- Convener

Dr. B.ANIL KUMAR

Convener

Dr. T. S. Kishore

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STEPCONE - 2024
ATTENDANCE SHEET

(5)

Department: EEE
Name of the Event: Technical Paper Presentation (Central Event)
Judge/Resource Person Details: Dr. G. Chandrasekhar, Dr. G. Indira Kishore

Date: 16th Feb. 2024
Venue: 03-S-03
Time: 11:00 AM TO 5:00 PM

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11.	Kodali Sriram	GMRIT	8328033114		
12.	V. Thrisha	GMRIT	7901250504		
13.	D. Thrivedi	GMRIT	8328337205		
14.		GMRIT			
15.		GMRIT			
16.	P. Pavan Sandilya	GMRIT	8074207441		
17.		GMRIT			

Event Coordinator

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STEPCONE - 2024
ATTENDANCE SHEET

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Name of the Event: Venue:
Judge/Resource Person Details: Time:

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19.	ch. Bhavani	GMRIT	8886850445	22341A0227@gmrit.edu.in	ch. Bhavani
20	v. Divya	GMRIT	9346812066	22341A04C4@gmrit.edu.in	V. Divya
21	S. Jahnavi	GMRIT	8328008939	2134IA0297@gmrit.edu.in	S. Jahnavi
22	SK. Nageena	GMRIT	6301211591	22345A0215@gmrit.edu.in	SK. Nageena
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Event Coordinators

GMR Institute of Technology
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Report of the Event (Paper Presentation)
16/02/2024

The Project and Innovation Club of the Department of Electrical and Electronics Engineering proudly organized the Paper Presentations event on February 16th, 2024, from 11:00 AM to 5:00 PM, with 26 participants in attendance. This event offered an intellectually stimulating experience where participants had the opportunity to present their research findings, innovative ideas, and solutions. It provided a platform for showcasing academic prowess, engaging with like-minded individuals, and inspiring discussions on cutting-edge topics. The event emphasized intellectual exchange and knowledge dissemination, allowing participants to let their ideas resonate in an environment dedicated to scholarly dialogue.

Signature of the HoD
Dr. Pilla Ramana
Place: GMR IT, Rajam